

HDOC Day-22

Write a menu – driven C program that accepts two positive integers, m and n , where $m \leq n$. Display the following menu and use a switch statement to perform the operation selected by the user.

- **Print the multiplication tables of m and n by side, using multipliers from 1 to 10.**
- **Display a multiplication grid containing the products of integer from 1 to m and integers from 1 to n .**
- **For each integer from m to n , print that integer followed by the next five consecutive integers.**
- **Exit the program.**

Algorithm:

Step 1: Start.

Step 2: Input two positive integers m and n .

Step 3: Check that $m \leq n$. If not, display an error message and take the input again.

Step 4: Display the menu:

Multiplication tables of m and n

Multiplication grid

Consecutive integers

Exit

Step 5: Input the user's choice.

Step 6: Perform the selected operation:

Choice 1:

- Run a loop from $i = 1$ to 10.
- Calculate $m \times i$ and $n \times i$.
- Display both tables side by side.

Choice 2:

- Print column headings from 1 to n.
- For every row from 1 to m, calculate row × column.
- Display the multiplication grid.

Choice 3:

- Run a loop from i = m to n.
- For each i, print i followed by i+1, i+2, i+3, i+4, i+5.

Choice 4:

- Display an exit message.
- Terminate the program.
- Invalid Choice
- Display "Invalid choice! Please enter a valid choice."
- Display the menu again.

Step 7: Repeat Steps 4–6 until the user chooses Exit.

Step 8: Stop.

C Program:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int m, n, choice, i, j;
```

```
    // Input two positive integers
```

```
    do
```

```
    {
```

```
        printf("Enter two positive integers m and n (m <= n): ");
```

```
        scanf("%d %d", &m, &n);
```

```
        if (m <= 0 || n <= 0 || m > n)
```

```
        {
```

```

        printf("Invalid input! Enter positive integers with m <= n.\n");
    }

} while (m <= 0 || n <= 0 || m > n);

// Menu-driven program
do
{
    printf("\n1. Multiplication tables of m and n\n");
    printf("2. Multiplication grid\n");
    printf("3. Consecutive integers\n");
    printf("4. Exit\n");

    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            // Multiplication tables of m and n
            printf("\nMultiplication Tables of %d and %d\n", m, n);

            for (i = 1; i <= 10; i++)
            {
                printf("%d x %d = %-3d   %d x %d = %-3d\n",
                    m, i, m * i, n, i, n * i);
            }
            break;

```

case 2:

```
// Multiplication grid
```

```
printf("\nMultiplication Grid (m=%d, n=%d)\n", m, n);
```

```
printf("  ");
```

```
for (j = 1; j <= n; j++)
```

```
{
```

```
    printf("%4d", j);
```

```
}
```

```
printf("\n");
```

```
printf("  ");
```

```
for (j = 1; j <= n; j++)
```

```
{
```

```
    printf("----");
```

```
}
```

```
printf("\n");
```

```
for (i = 1; i <= m; i++)
```

```
{
```

```
    printf("%2d |", i);
```

```
    for (j = 1; j <= n; j++)
```

```
    {
```

```
        printf("%4d", i * j);
```

```
    }
```

```
    printf("\n");  
}  
break;
```

case 3:

```
// Consecutive integers  
printf("\nConsecutive Integers from %d to %d (m=%d, n=%d)\n",  
    m, n, m, n);  
  
for (i = m; i <= n; i++)  
{  
    printf("%d:", i);  
  
    for (j = 0; j <= 5; j++)  
    {  
        printf(" %d", i + j);  
    }  
  
    printf("\n");  
}  
break;
```

case 4:

```
// Exit  
printf("\nExiting the program...\n");  
break;
```

default:

```

        // Invalid choice

        printf("\nInvalid choice! Please enter a choice between 1 and 4.\n");

    }

} while (choice != 4);

return 0;
}

```

Test Case Table:

Test Case	Input m,n	Choice	Expected Output	Result
1	3,5	1	Tables of 3 and 5 from *1 to *10	Pass
2	3,5	2	Grid from 1 to 3* 1 to 5	Pass
3	3,5	3	Consecutive integers from 3 to 5	Pass
4	3,5	4	Program exits	Pass
5	3,5	7	Invalid choice message	Pass
6	5,3	-	Invalid input because $m > n$	Pass
7	-2,5	-	Invalid input because m is not positive	Pass

Compilation and Execution:

The following program has been compiled and executed successfully.

Enter two positive integers m and n ($m \leq n$)

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 1

Multiplication Tables of 3 and 5

3 * 1 = 3	5 * 1 = 5
3 * 2 = 6	5 * 2 = 10
3 * 3 = 9	5 * 3 = 15
3 * 4 = 12	5 * 4 = 20
3 * 5 = 15	5 * 5 = 25
3 * 6 = 18	5 * 6 = 30
3 * 7 = 21	5 * 7 = 35
3 * 8 = 24	5 * 8 = 40
3 * 9 = 27	5 * 9 = 45
3 * 10 = 30	5 * 10 = 50

Multiplication Grid (m=3, n=5)

	1	2	3	4	5
1	1	2	3	4	5
2	2	4	6	8	10
3	3	6	9	12	15

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 3

Consecutive Integers from 3 to 5 (m=3, n=5)

3: 3 4 5 6 7 8
4: 4 5 6 7 8 9
5: 5 6 7 8 9 10

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 4

Exiting program...

(kali@kali)-[~/Desktop/exp22]
\$./exp22

Enter two positive integers m and n ($m \leq n$): 5 3
Invalid input. Enter positive integers with $m \leq n$.
Enter two positive integers m and n ($m \leq n$): -2 5
Invalid input. Enter positive integers with $m \leq n$.
Enter two positive integers m and n ($m \leq n$): 5 5

```

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 1

Multiplication Tables of 5 and 5 (m=5, n=5)

5 x 1 = 5      5 x 1 = 5
5 x 2 = 10     5 x 2 = 10
5 x 3 = 15     5 x 3 = 15
5 x 4 = 20     5 x 4 = 20
5 x 5 = 25     5 x 5 = 25
5 x 6 = 30     5 x 6 = 30
5 x 7 = 35     5 x 7 = 35
5 x 8 = 40     5 x 8 = 40
5 x 9 = 45     5 x 9 = 45
5 x 10 = 50    5 x 10 = 50

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 4

Exiting program...

(kali@kali)-[~/Desktop/exp22]
$ ./exp22
Enter two positive integers m and n (m ≤ n): 1 1

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit
Enter your choice: 2

```

Multiplication Grid (m=1, n=1)

```
  1
  --
1 | 1
```

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 4

Exiting program...

```
(kali@kali)-[~/Desktop/exp22]
$ ./exp22
```

Enter two positive integers m and n ($m \leq n$): 3 5

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 7

Invalid menu choice. Please enter a choice from 1 to 4.

1. Multiplication tables of m and n
2. Multiplication grid
3. Consecutive integers
4. Exit

Enter your choice: 4

Exiting program...